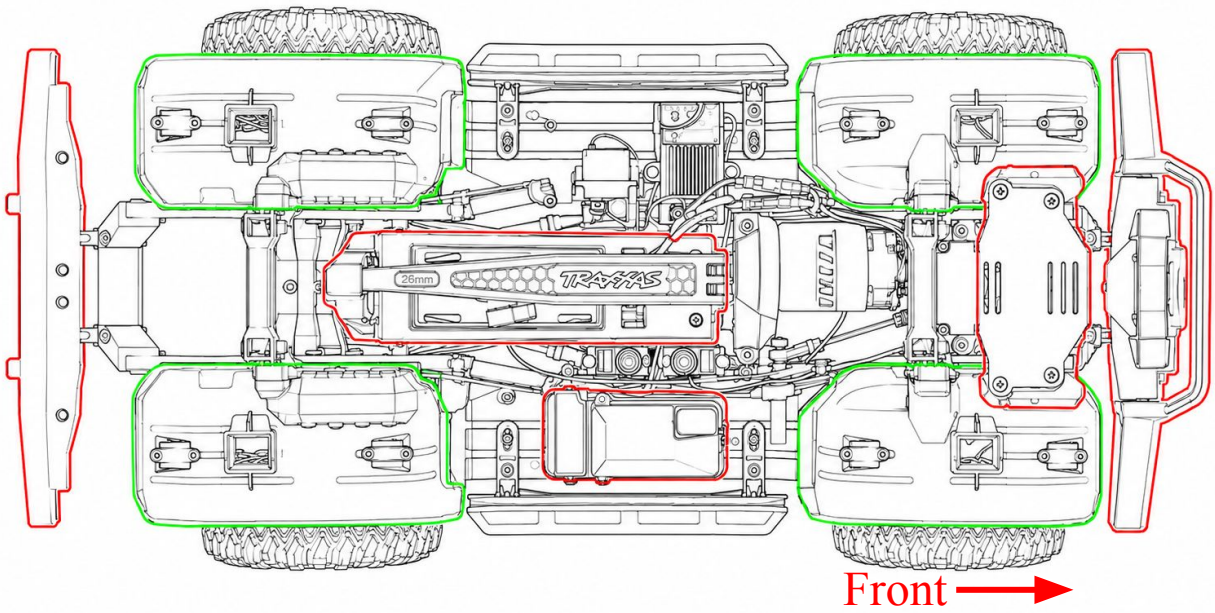
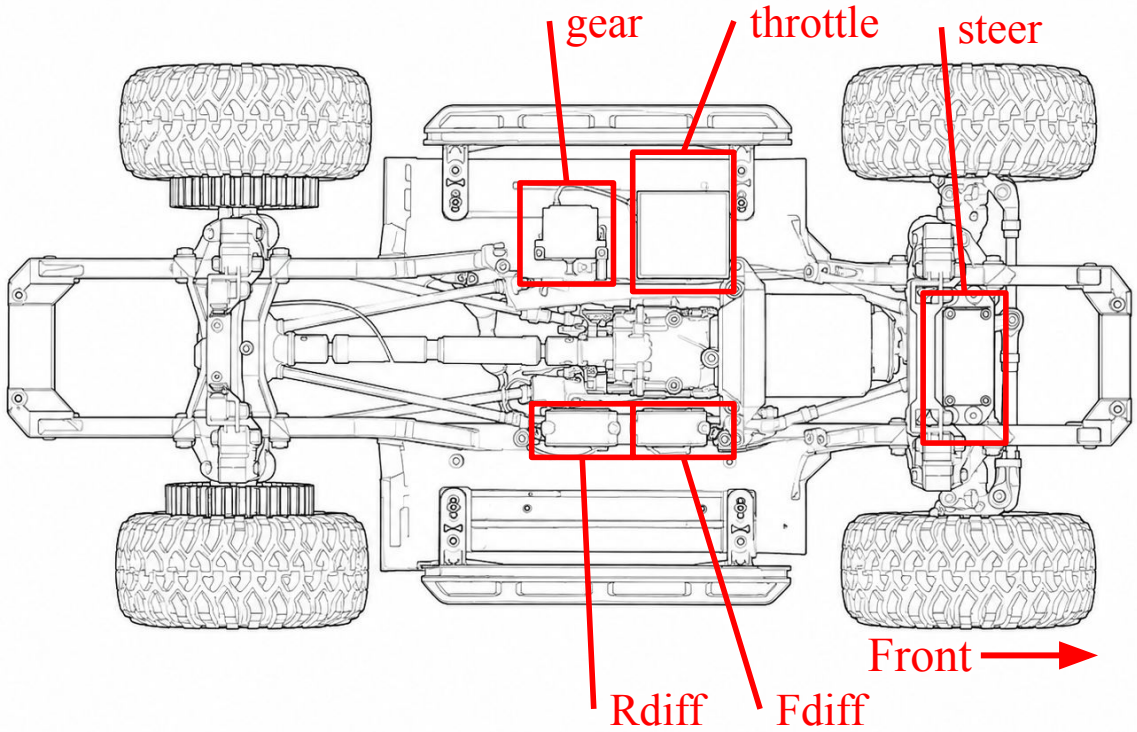
Red: Remove Parts
Green: Switch the fenders over the car wheels to shock towers
https://traxxas.com/8216-trx-4-front-and-rear-shock-towers-lr

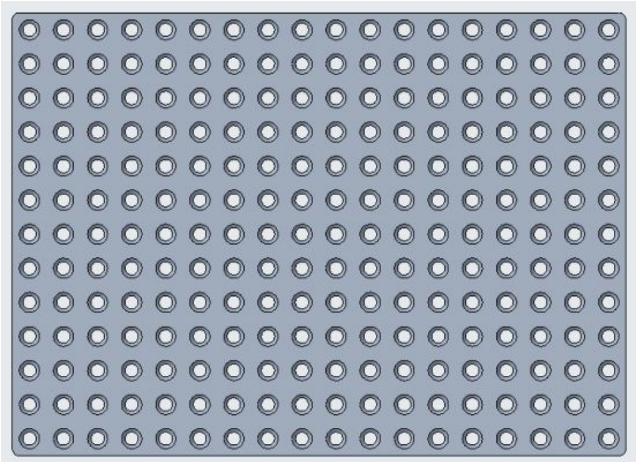


Result should look like the following image

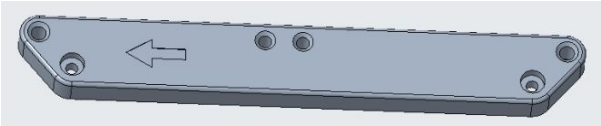
Remember to label all the cables



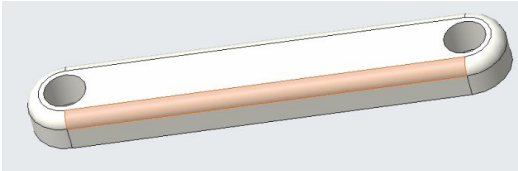
svea_plate_half x 2



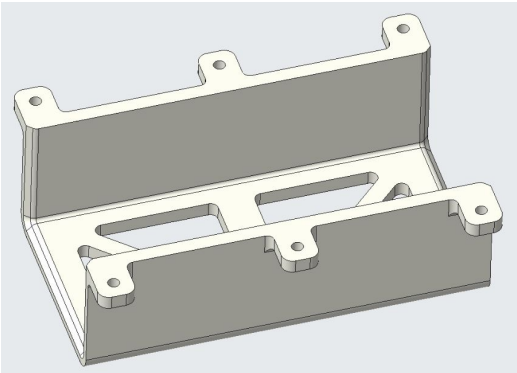
svea_plate_mount x 2



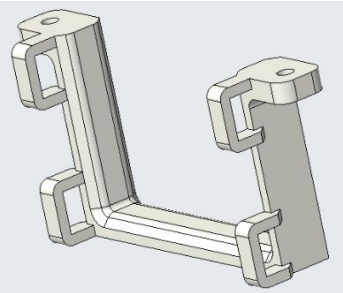
svea_battery_stopper x 1



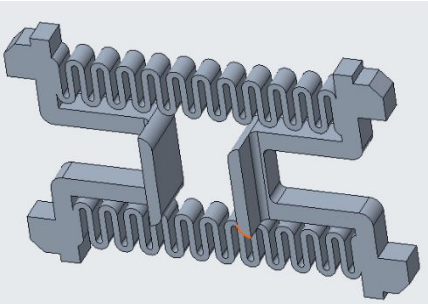
svea_battery_holder x 1



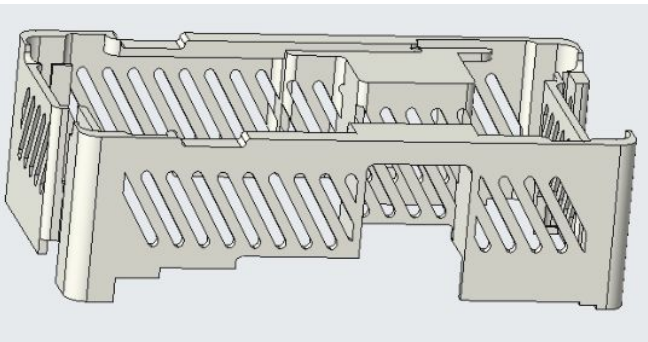
svea_battery_back x 1



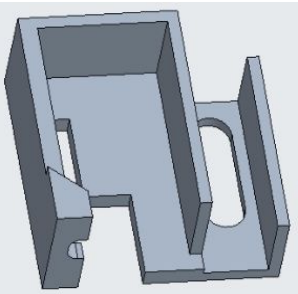
svea_battery_back_cover x 1



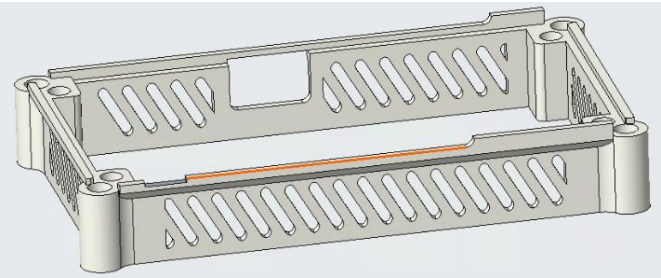
svea_pmb_cover x 1



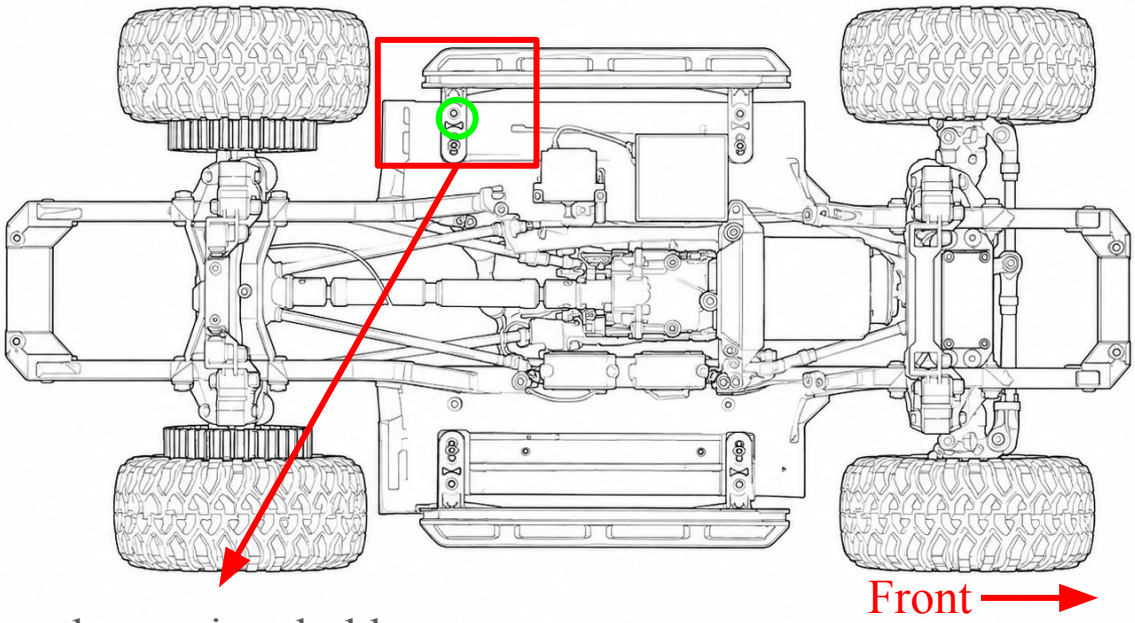
svea_receiver_holder x 1



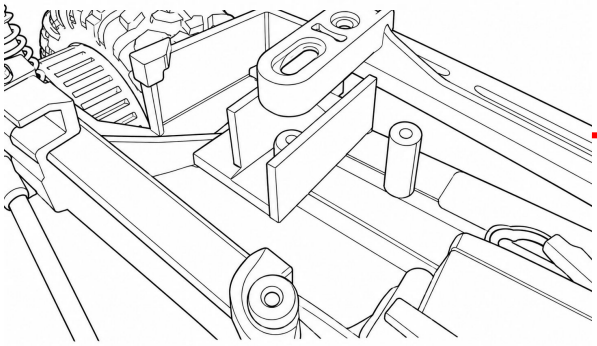
svea_pmb_stand x 1



Remove the bolt in the Green circle temporarily

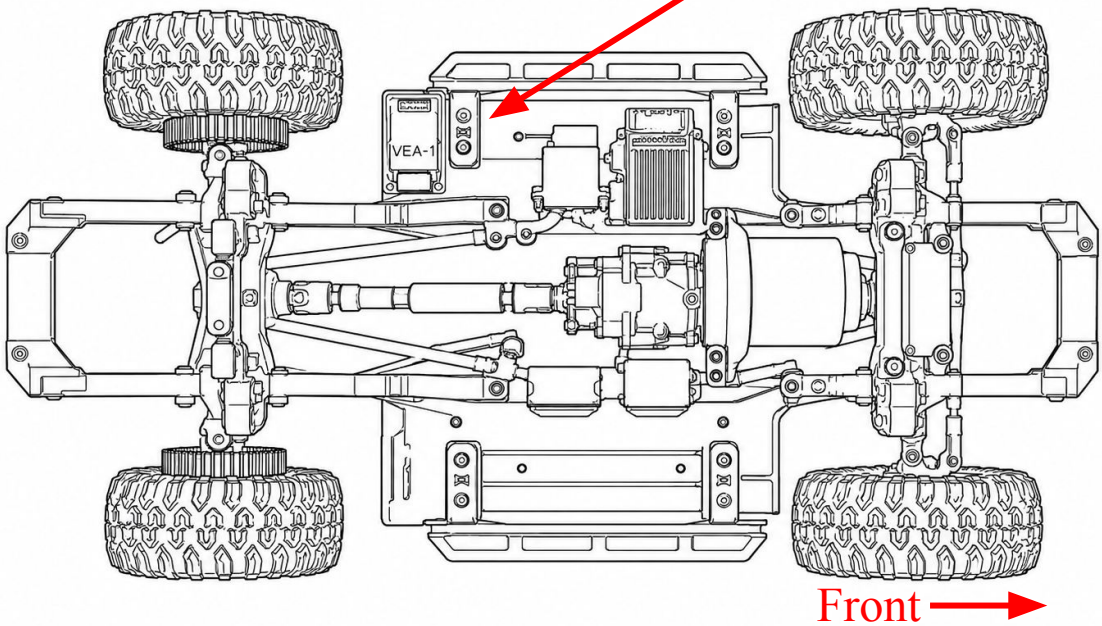


Place the receiver holder underneath the bar and put the screw back.

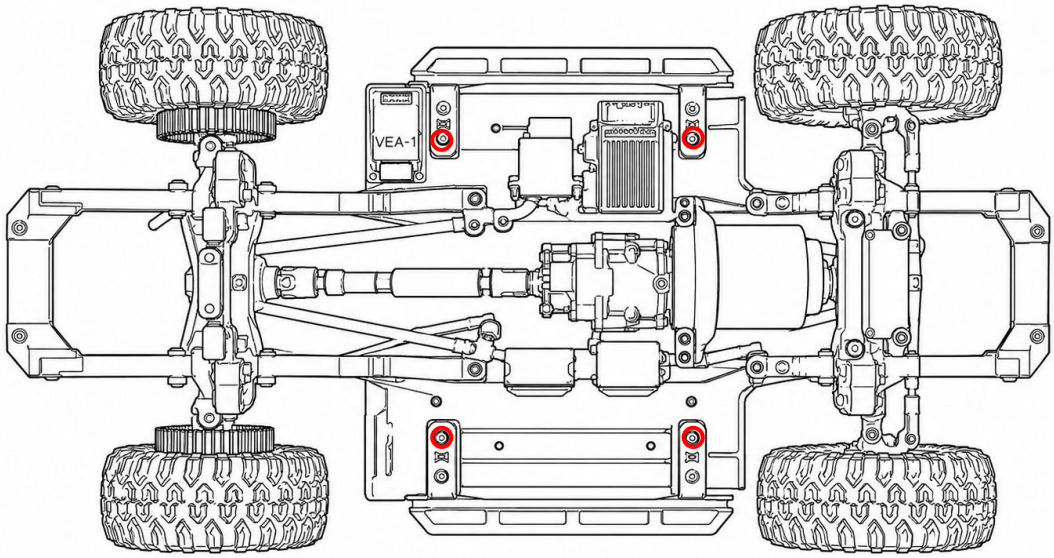


Place the receiver in the holder.





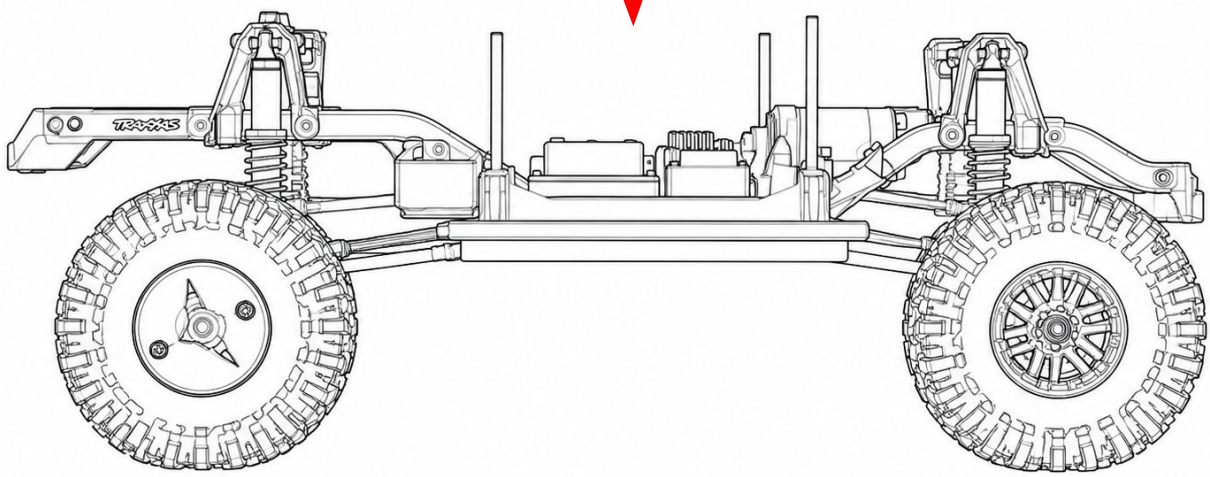
Thread the M3×60mm standoffs into the four threaded holes marked with red circles.



Front

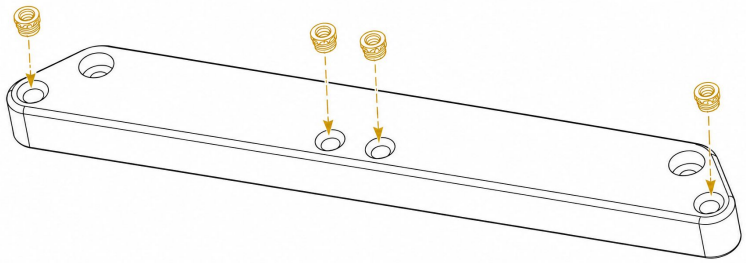
M3×60mm standoffs * 4



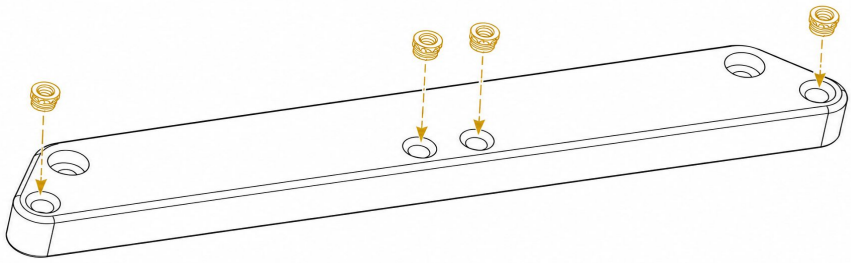


Front

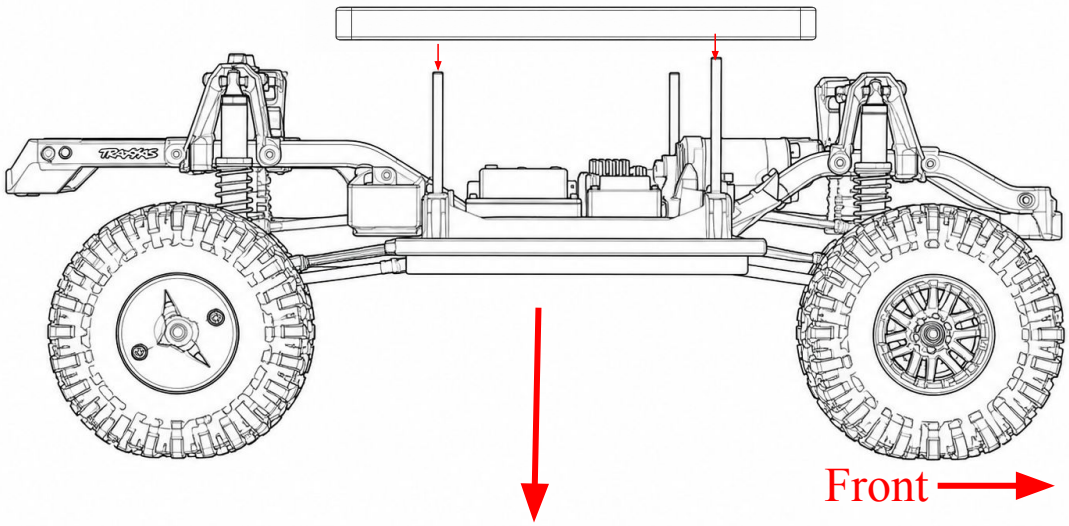
Insert 4 M3 heated inserts to the plate mount.



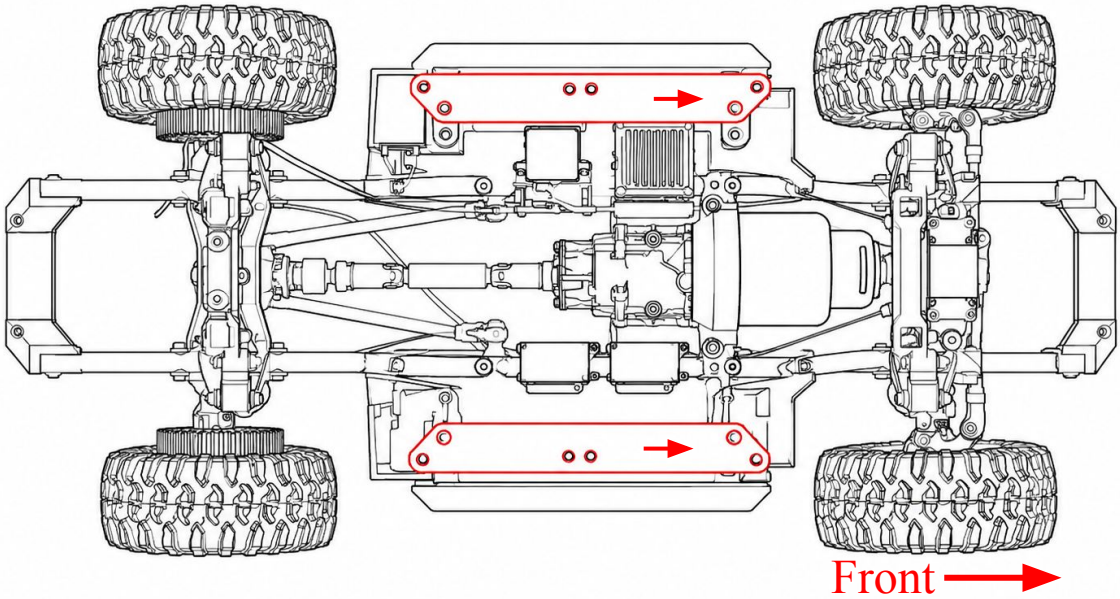
Insert 4 M3 heated inserts to another plate mount on the mirror side.



Install plate mount with M3*10mm socket head screws.



Shorter side should be toward the back.

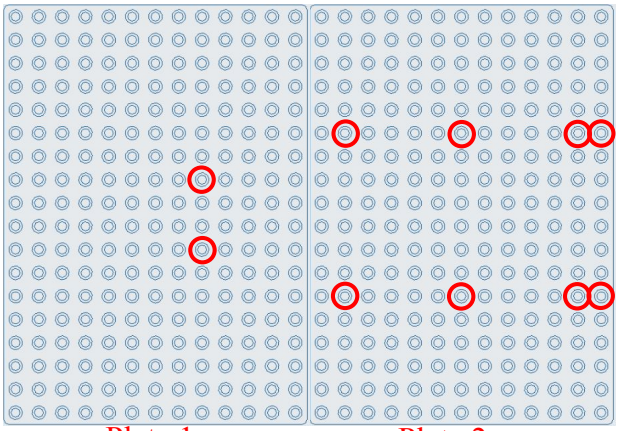


Recommended M3 heat inserts placement

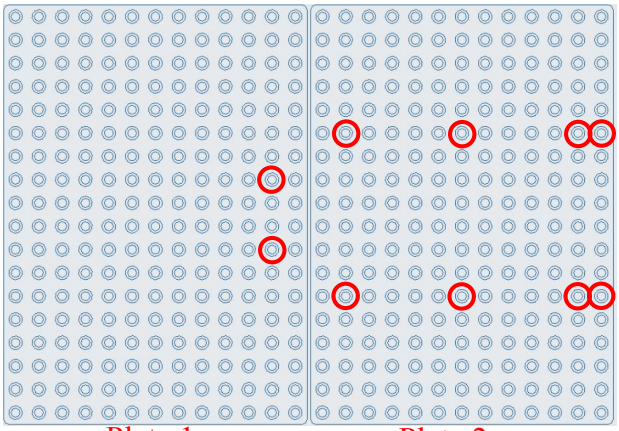
Front

Bottom

For 9000mAh Battery

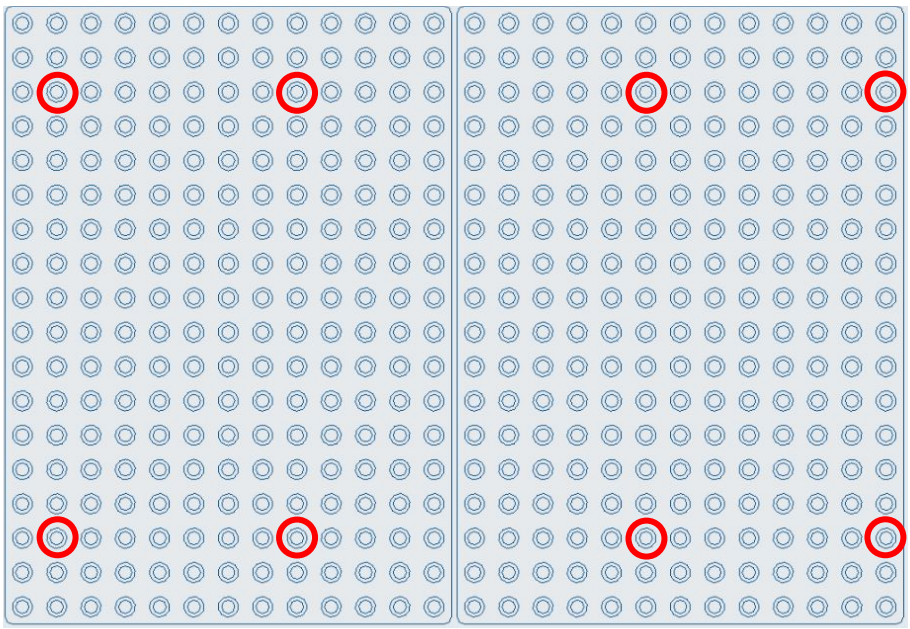


For 8000mAh Battery



Front

Top



Use M3*6mm or M3*10mm Screws to install battery holder.

Plate 1 Bottom

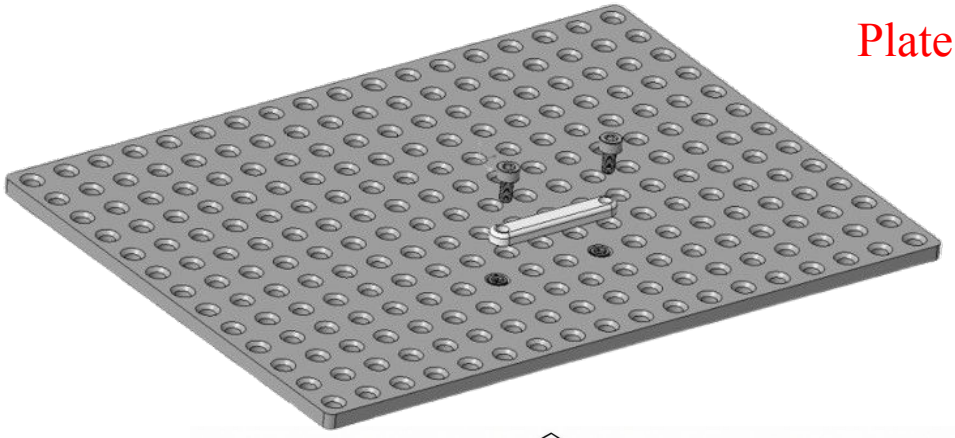
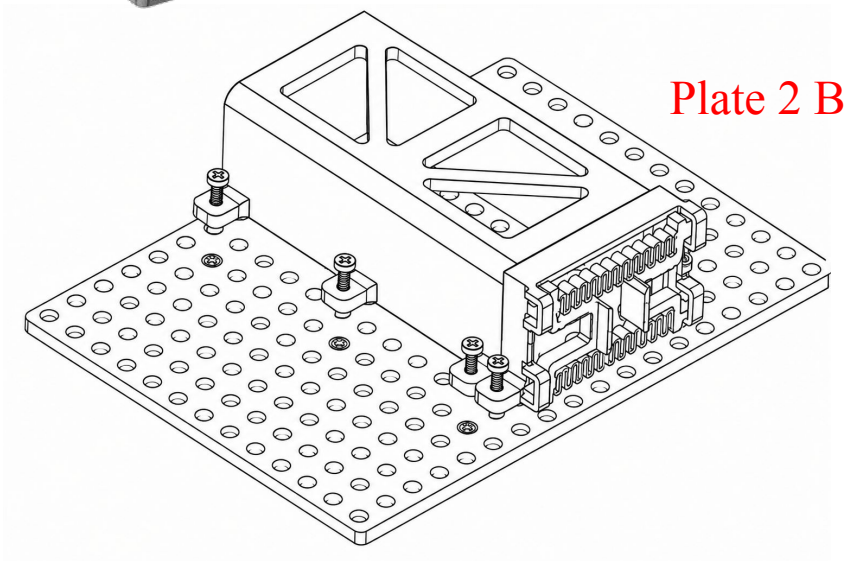
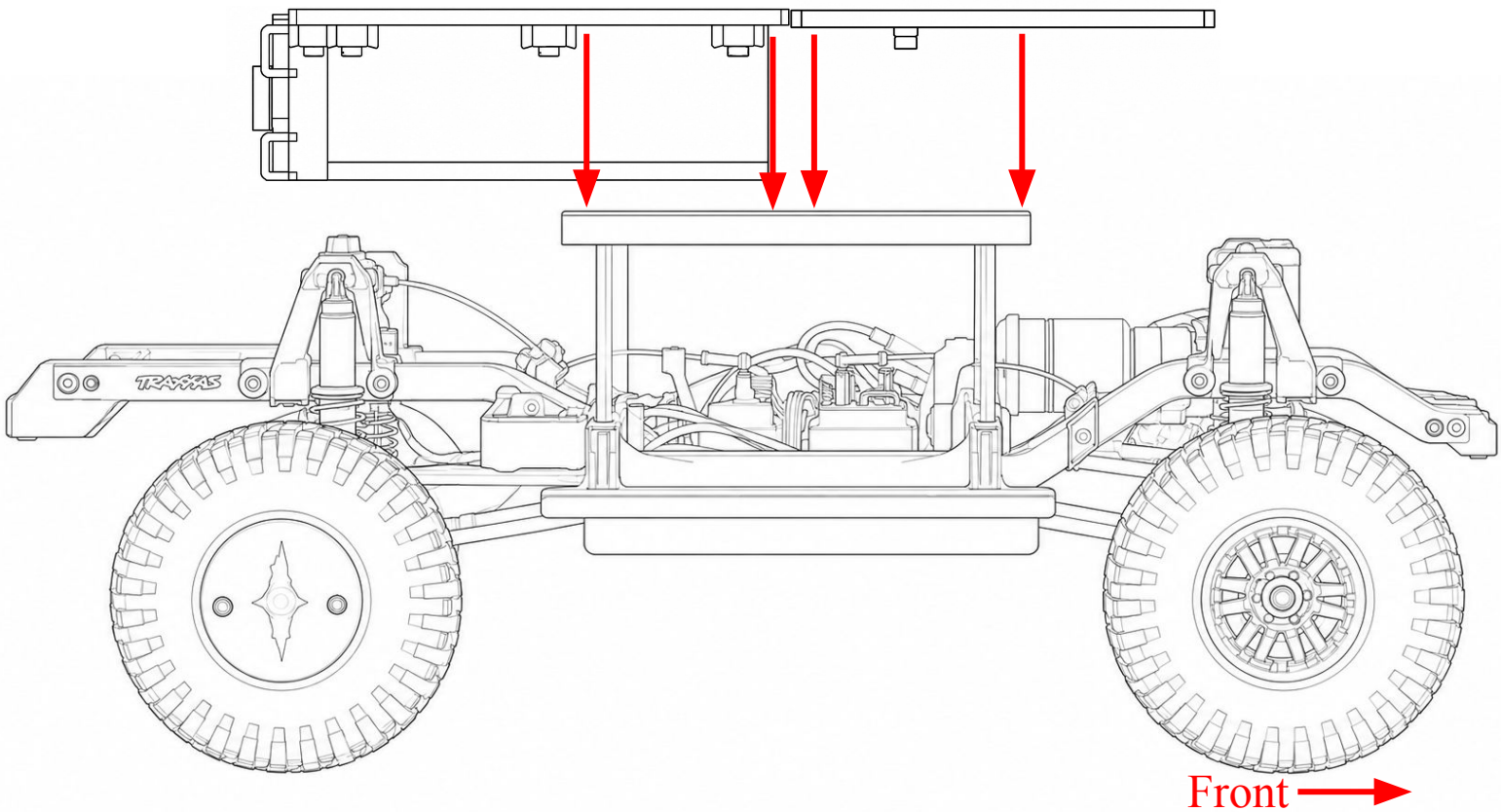


Plate 2 Bottom

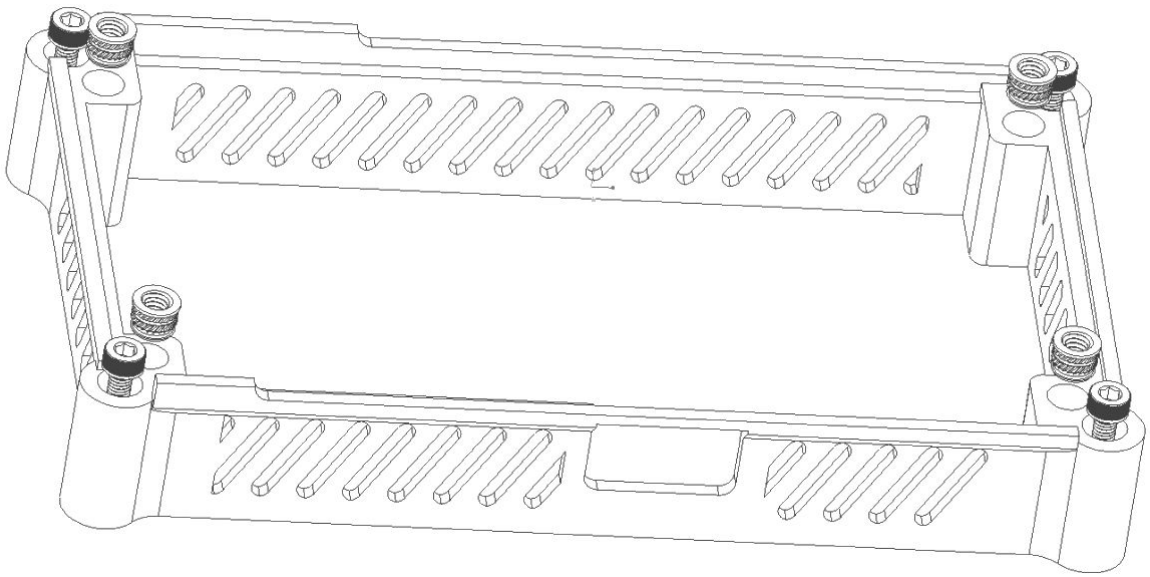


Use M3*10mm sink head screws to install plates.

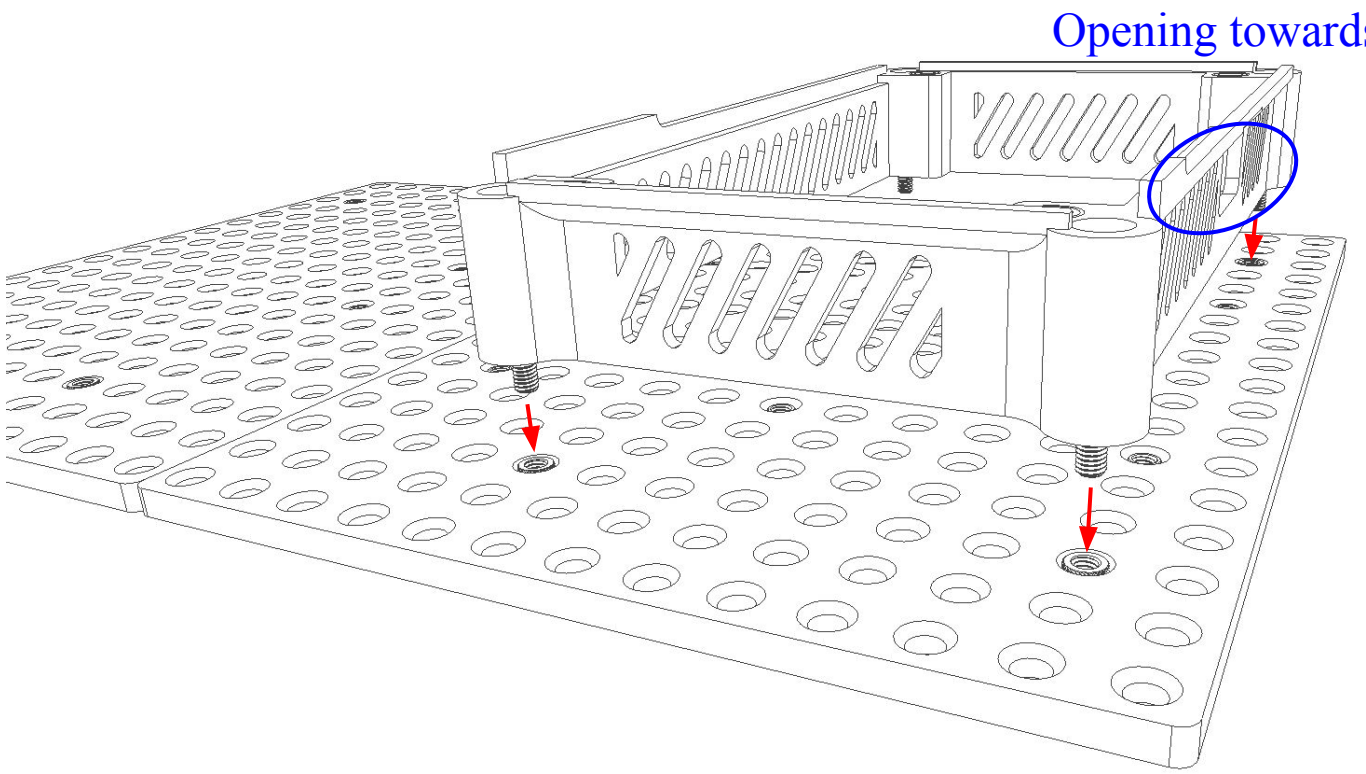
*Sort cables to left back of the chassy before install the plate.



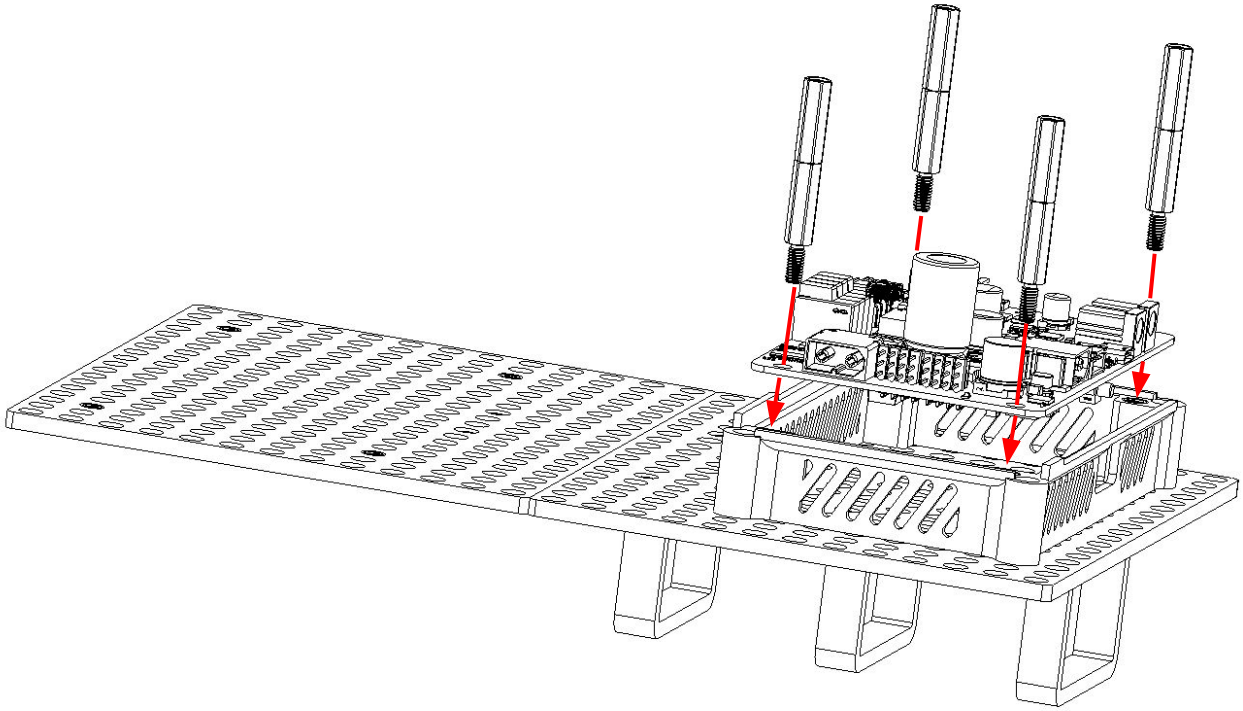
Insert M3*10mm screws to the outer holes and insert the M4*4.7mm heated insert to 4 inner holes.

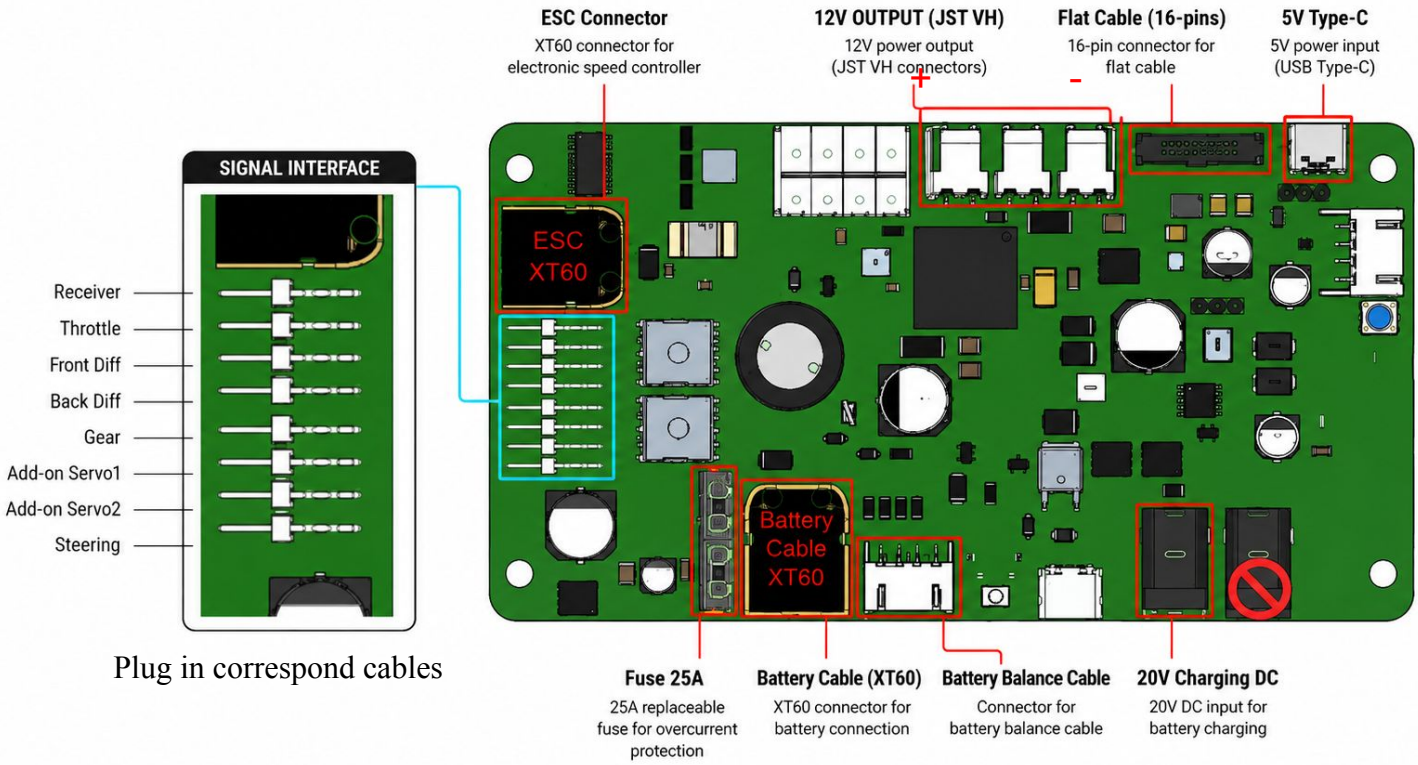


Install lower shell to the back of the board.



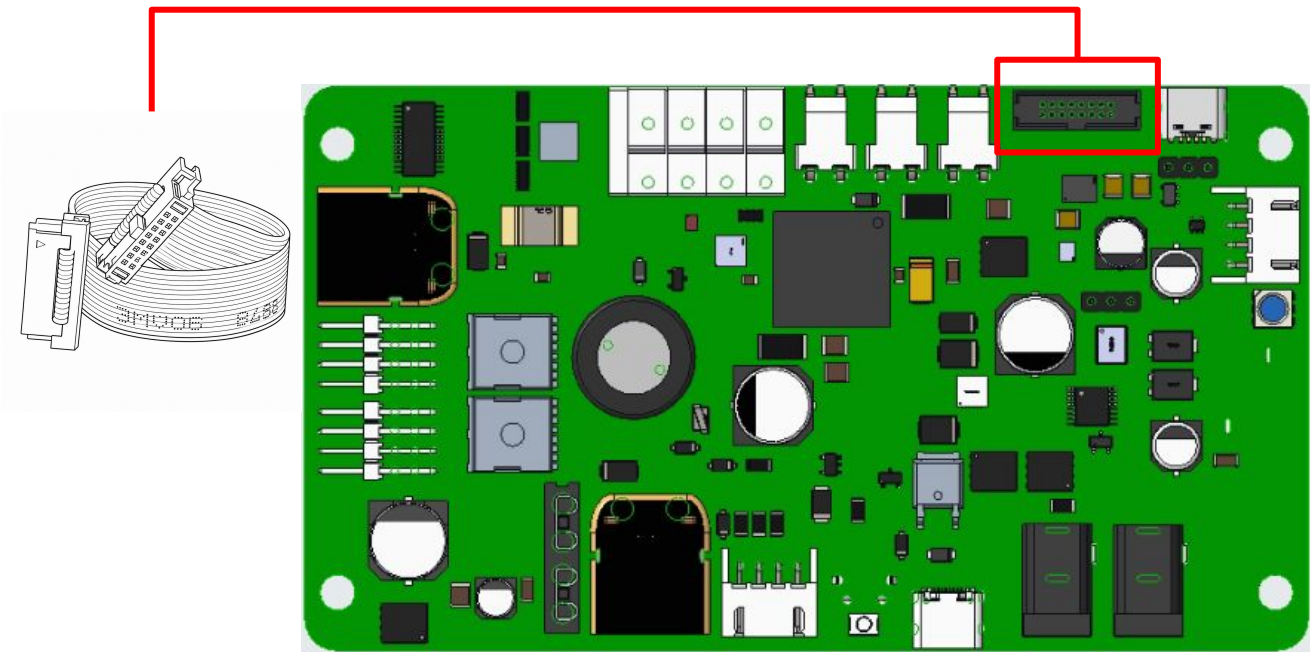
Install SVEA powerboard on the lower shell with M4 standoffs (preferably 40mm in total length)



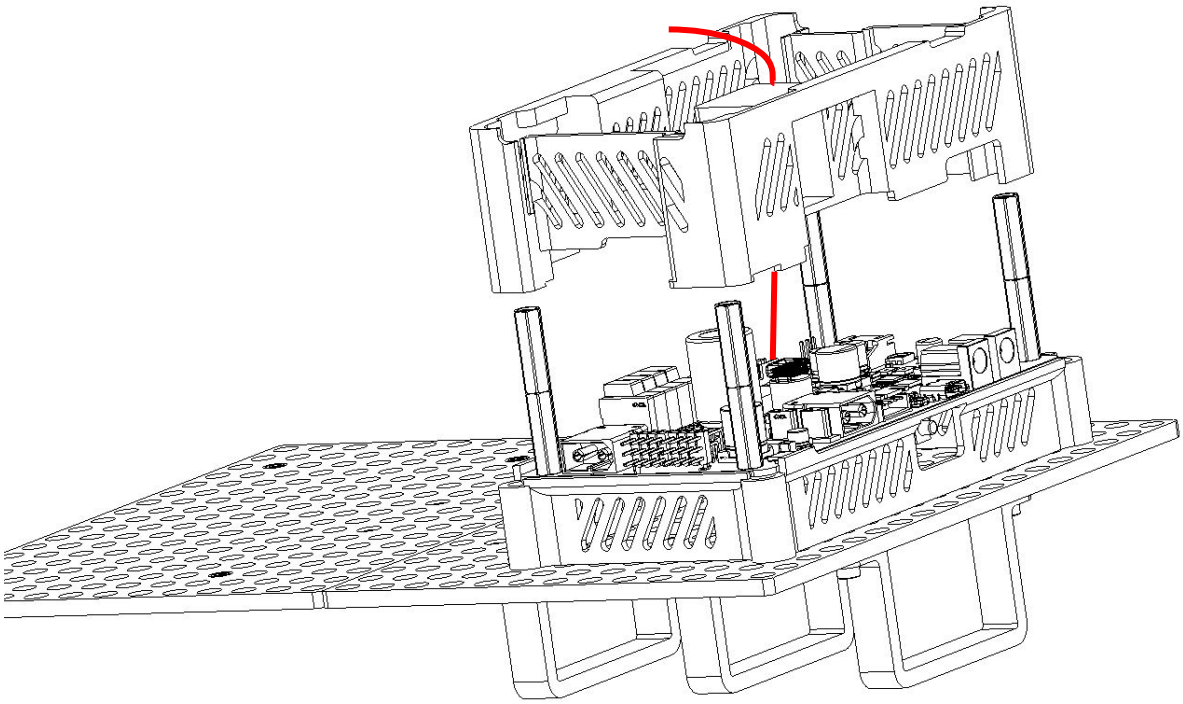


Plug in correspond cables

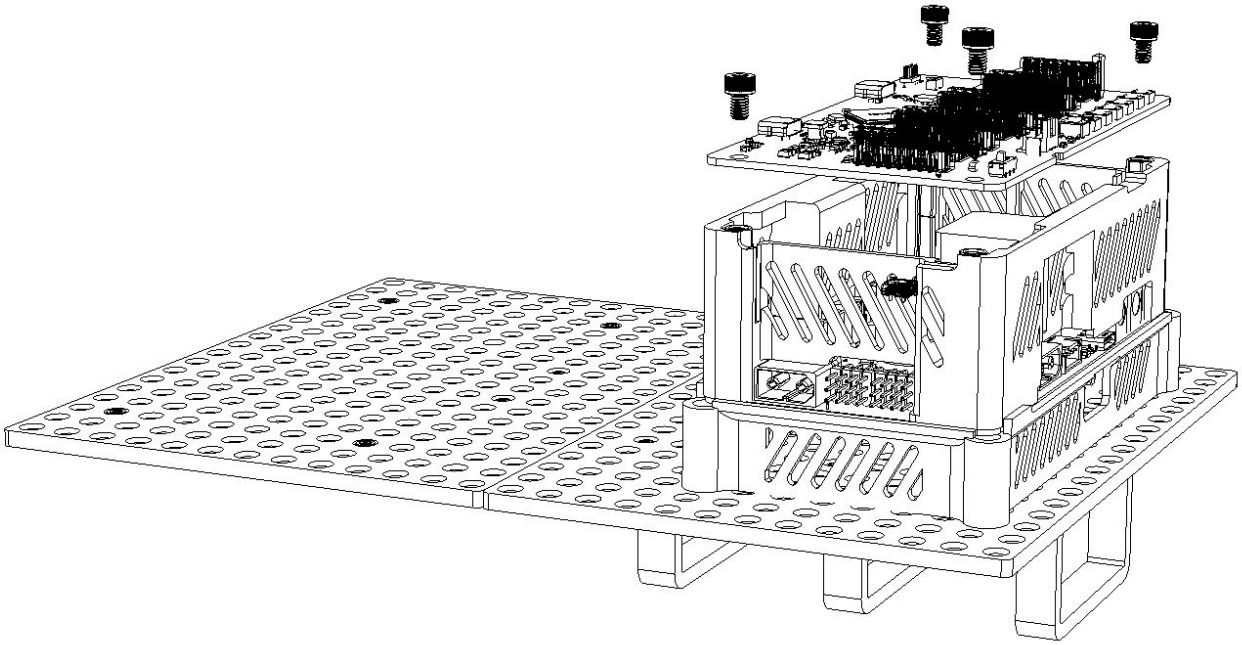
Plug the flat cable to PMB.
Flat Cable: link



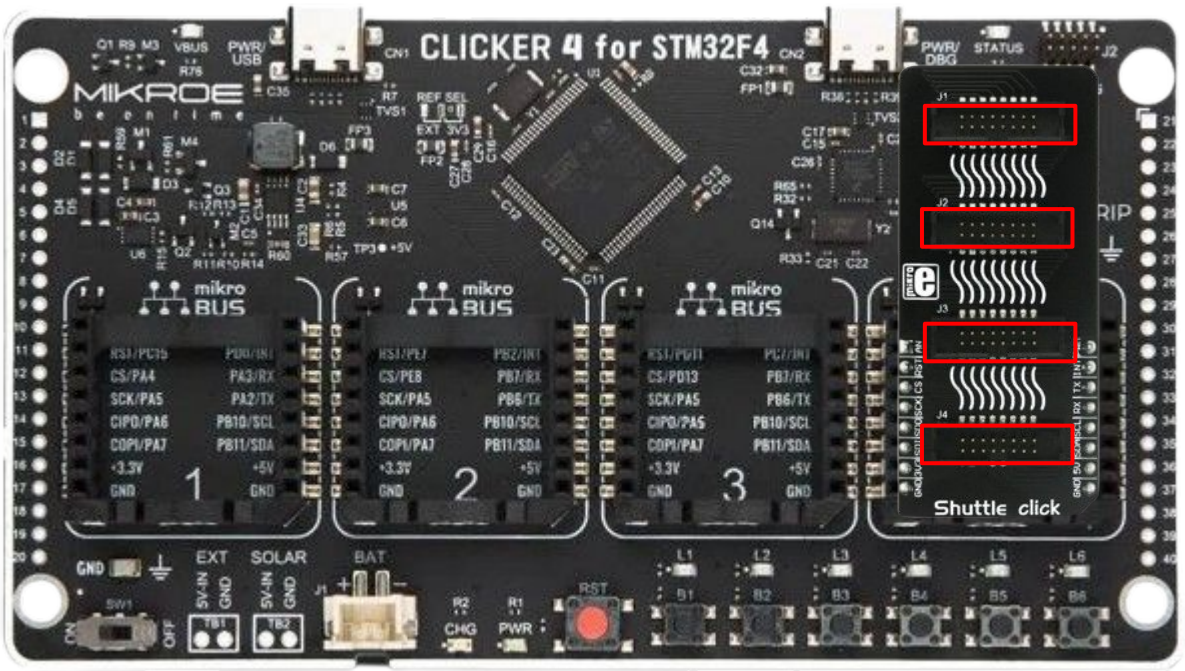
Slide the upper cover case in, pass the flat cable through the cover case.



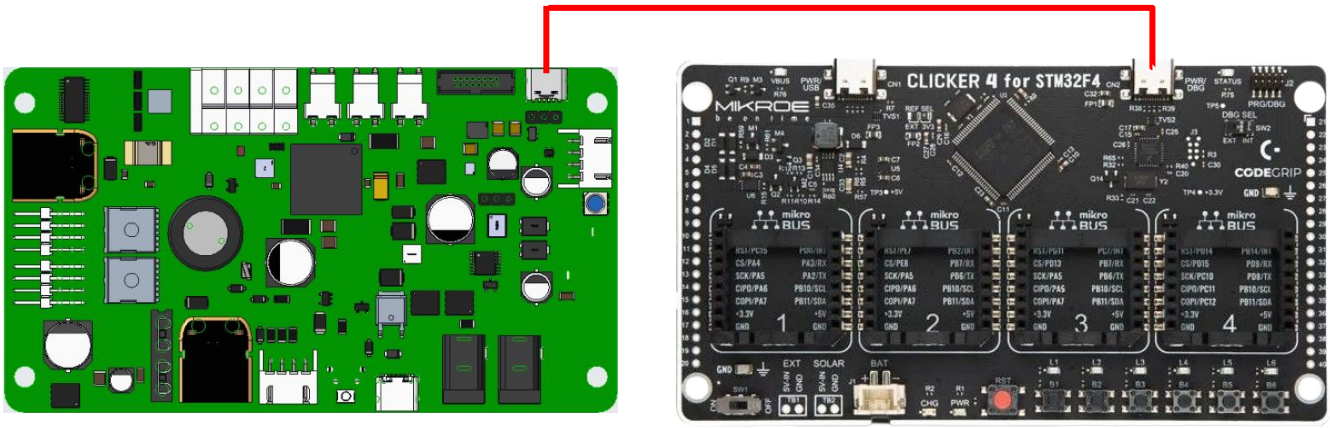
Install Clicker 4 on the standoffs with M4*6mm or M4*10mm screws. Flash the Clicker 4 with the following repo: https://github.com/nilskiefer/SVEA-PX4-Autopilot



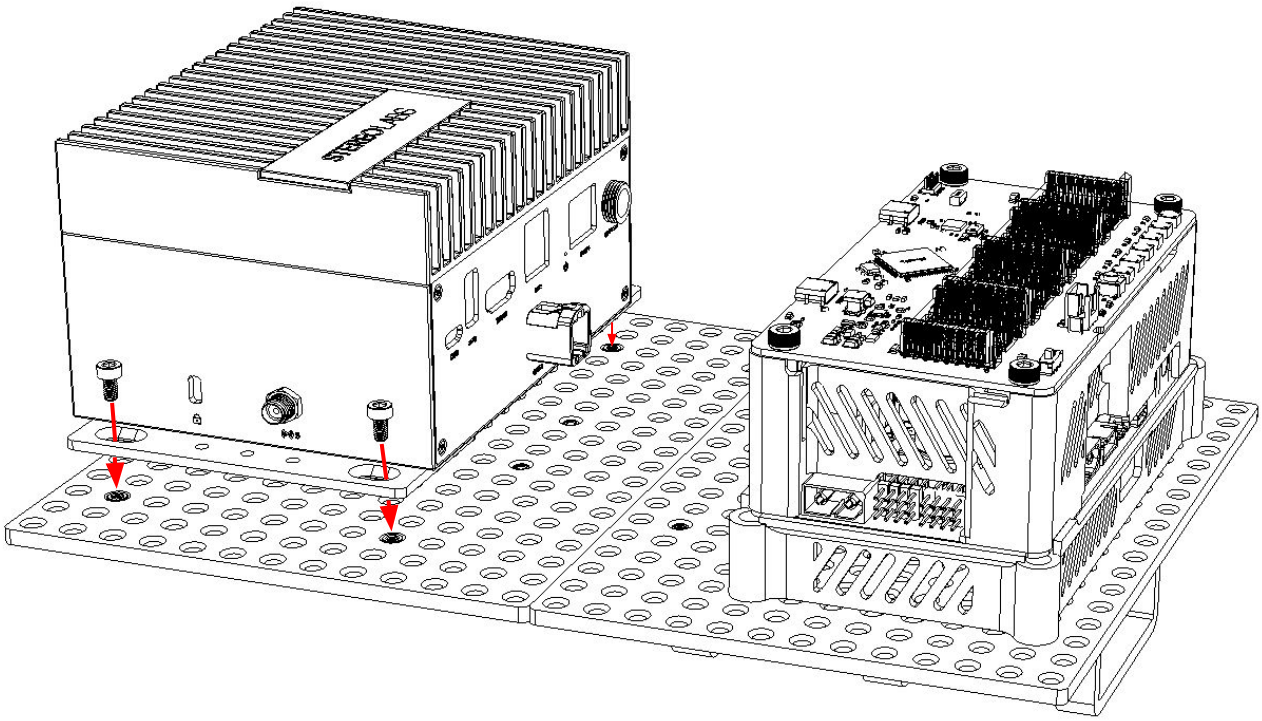
Plug the Shuttle Click to bus 4 on the Clicker 4. And plug flat cable to one of the port on the Shuttle Click (any port).



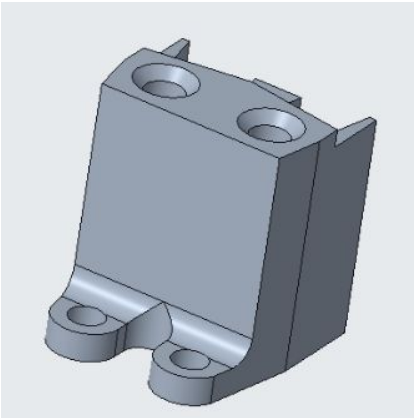
Power the clicker 4 with a type-c to type-c cable



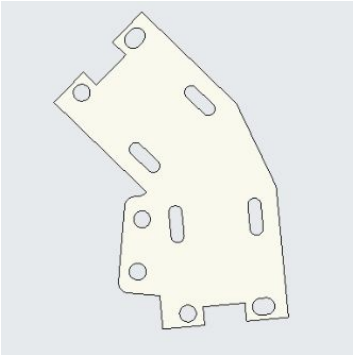
Install the zed box with 4 M3 screws



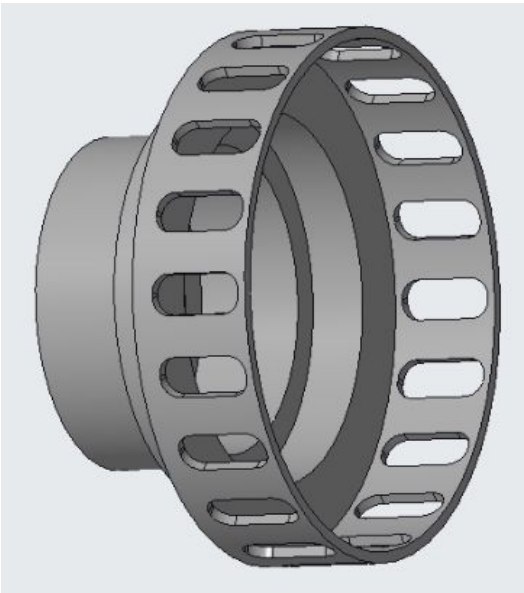
svea_duel_encoders_mount x 2



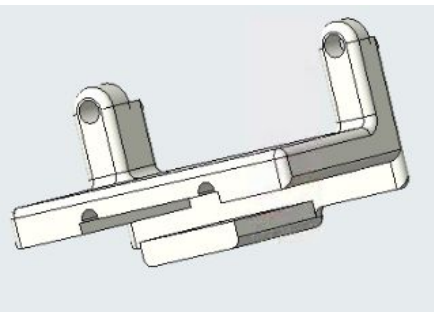
svea_duel_encoders_plate x 2



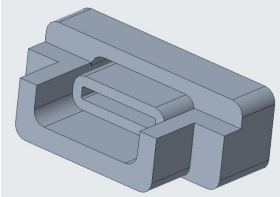
svea_wheel_encoder x 2



svea_esp32_mount x 1

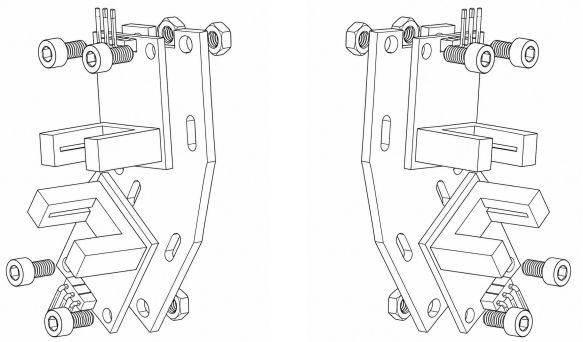


svea_esp32_type_c_plug x 1

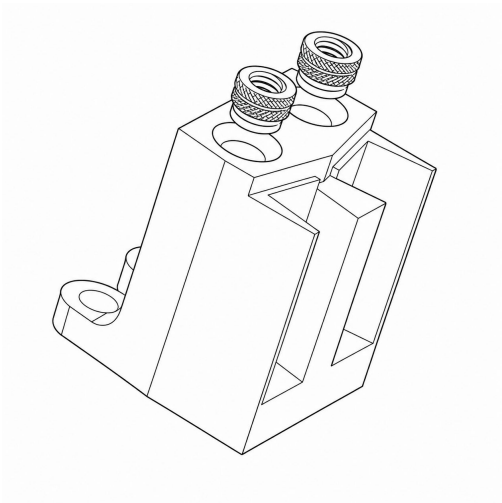


Encoder

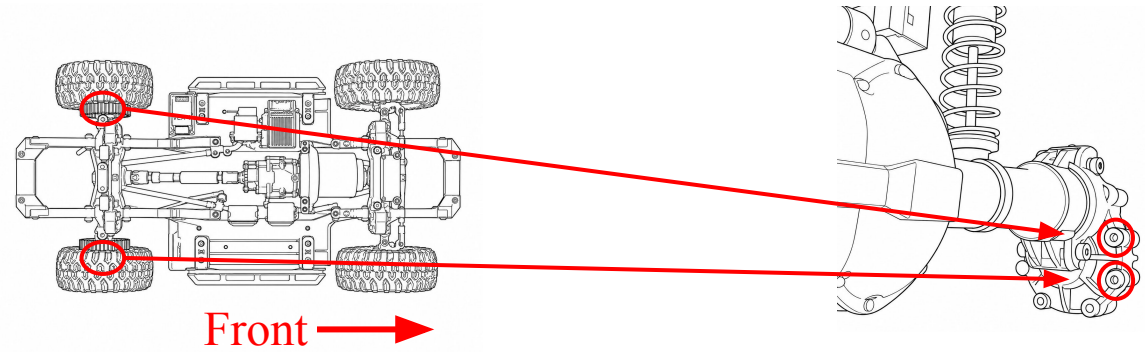
Install h2010 IR sensor to the encoder mount back cover with M3*6mm screws and M3 nuts.



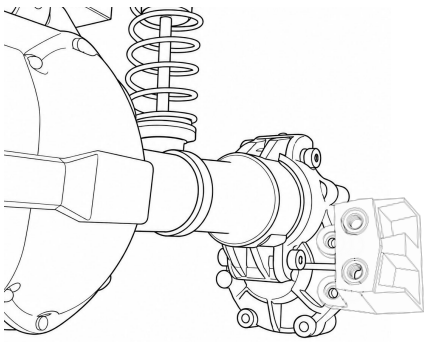
Insert the m3 heat insert to encoder mount.



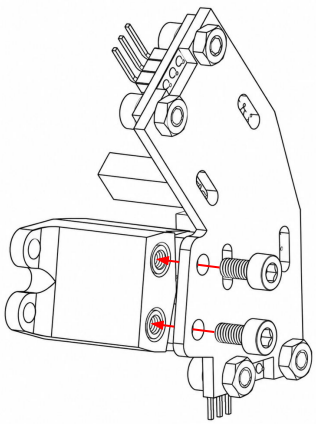
Remove these two screws for both sides.



Use the screws to install the encoder mount for both side

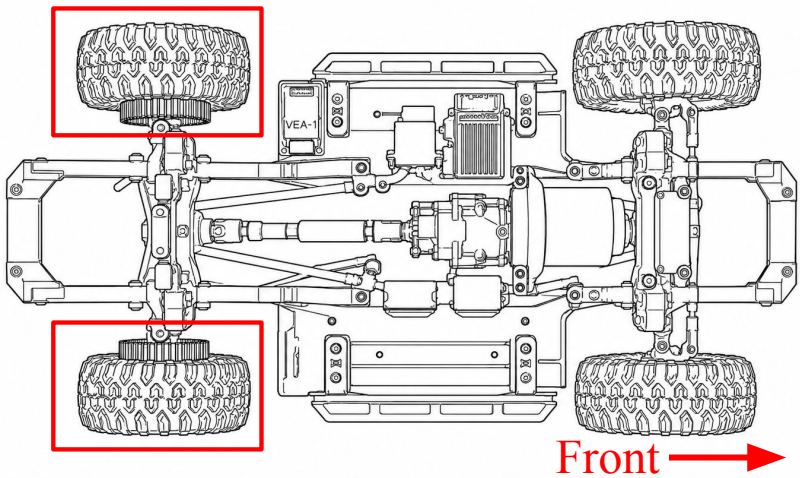


Install the sensor with M3*10mm

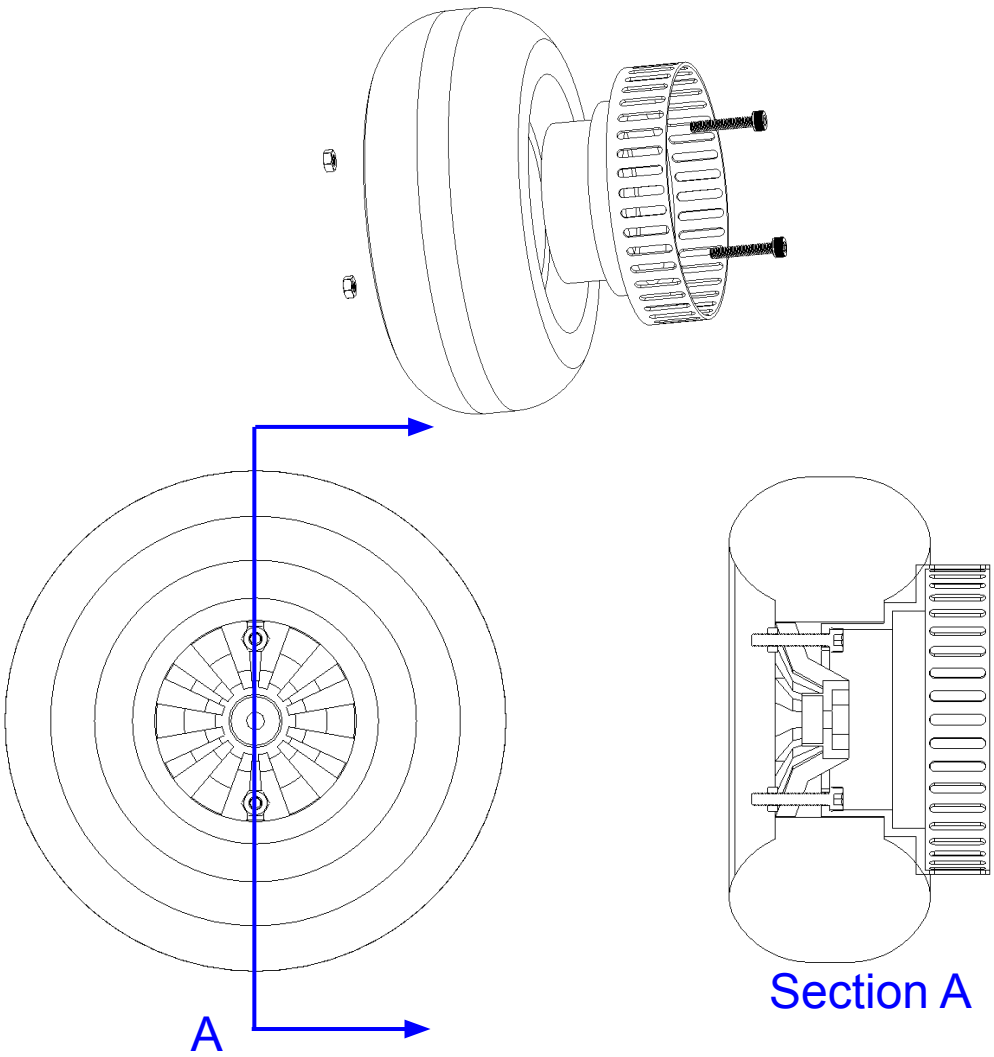


Encoder Disk

Remove the back two wheels.



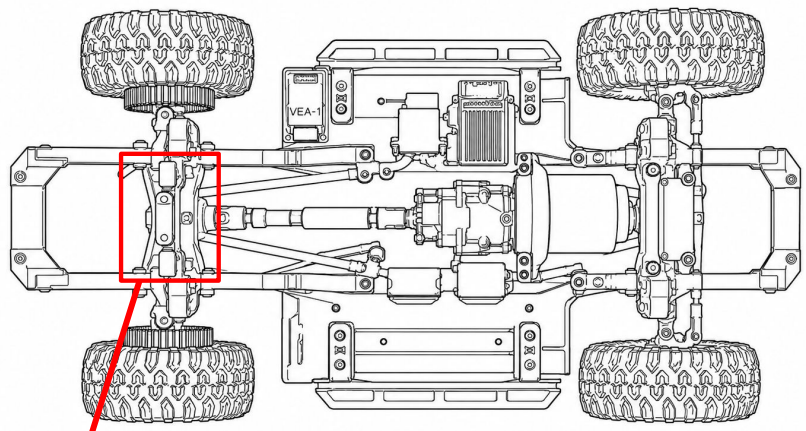
Install the wheel disk with two M3*18mm screws and 2 M3 nuts.



Put the wheel back.

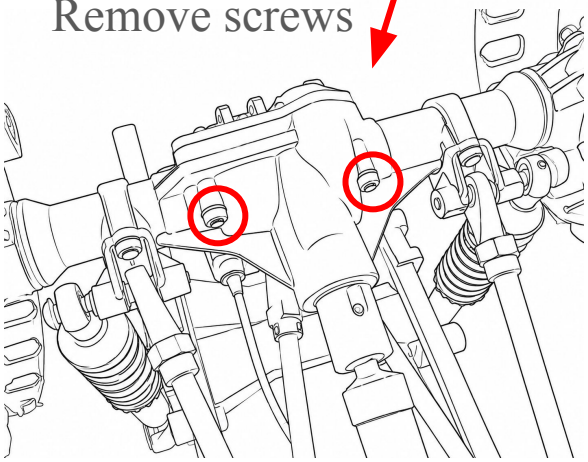
Encoder Chip ESP32

From the bottom of the chassy.

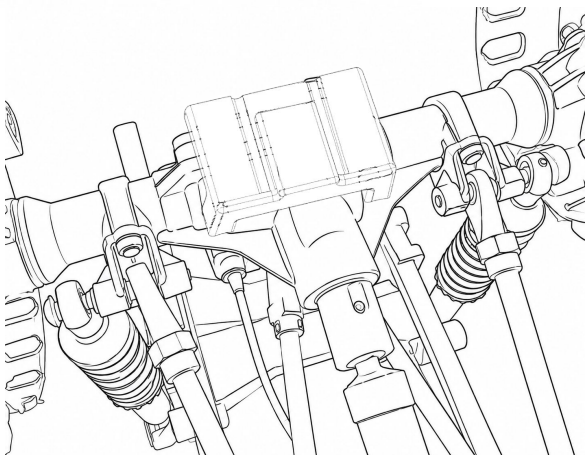


Front

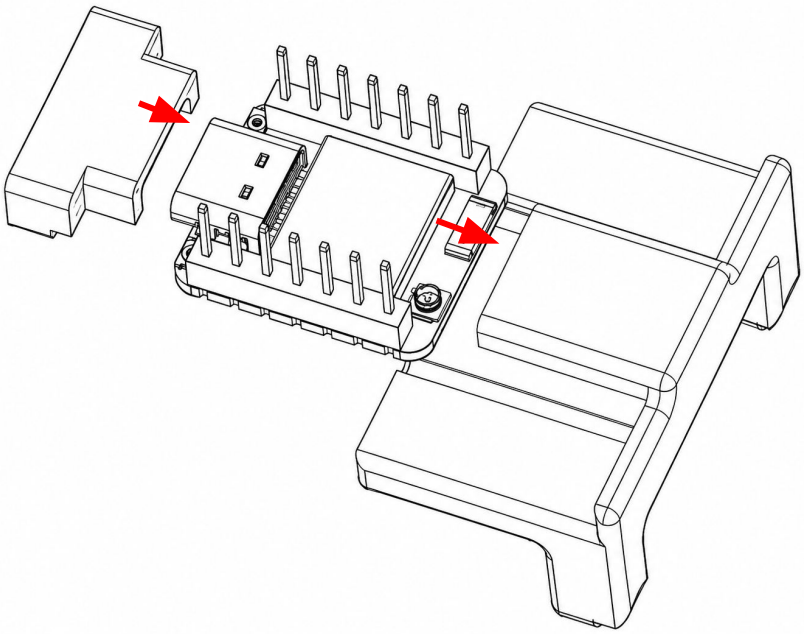
Remove screws



Use the screws to insert the holder



Use the screws to insert the holder,
flash chip with:
https://github.com/lnch223/SVEA-PX4-Autopilot-MCU-Peripheral/tree/main



Encoder Diagram

